ALPHA MICROSYSTEMS
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AlphaNET®
User's Guide

DSM-00063-03

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REVISIONS INCORPORATED	
REVISION	DATE
00	March 1989
01	July 1990
02	November 1991
03	February 1995

AlphaNET User's Guide
To re-order this document, request part number DSO-00063-00.

This document applies to AlphaNET versions 2.4 and later.

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This document was written by Robin Soto.

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CHAPTER 1 —INTRODUCTION AND OVERVIEW

In this chapter we discuss:

- What is a network?
- What is AlphaNET?
- What can AlphaNET do?
- Graphics conventions we'll be using

WHAT IS A NETWORK?

A network is a system linking together computers so messages, data, and files can be transferred from one computer to another. A network allows computers to talk to one another without the trouble of using transfer media such as magnetic tape or diskettes.

A network consists of the hardware necessary to link the computers together, plus the software that transfers the data and keeps track of the computers that are part of the network (knowing which ones are on and off and which ones are transmitting and receiving data).

The network software consists of a number of layers which are between the physical hardware and the operating system. These layers of software make it possible for a computer to be connected to more than one network at a time—in fact, to more than one kind of network at a time.

WHAT IS ALPHANET?

Alpha Micro has developed AlphaNET as a method of transferring files, exchanging information, and sharing resources between different Alpha Micro computers.

The AlphaNET software is used in conjunction with a choice of several physical interconnection methods, including asynchronous serial links (SerialNET) and Ethernet.

Each computer becomes a *node*—one of the connecting links on the network when it is connected to the AlphaNET system. Nodes are arranged in *groups*, which are user-defined subsections of a network. Group definitions give you the flexibility to organize and structure your network to meet your communication needs.

Multiple networks can be linked together using *gateways* which transparently connect networks so they can be treated as a single large network.

Because AlphaNET is a “layered”software product, it can take advantage of new interconnection methods without requiring changes to the way you use AlphaNET.

WHAT CAN ALPHANET DO?

You can use AlphaNET to:

- Transfer files to and from other computers
- Send messages to users on other computers
- Exchange information by electronic mail
- Get a directory of files on other computers
- Send files to printers on other computers for printing, so different computers can share printers.
- Connect your terminal to a remote computer, making it seem as if you had a direct connection.
- Get status information on remote computers.

AlphaNET is designed to make your use of remote computers completely transparent. You only need to specify the computer a file or user is on to be able to access it. AlphaNET takes care of all network routing.

A number of AMOS commands are not permitted to be used over the network because of the potential for harm to another computer. For example, you are not allowed to run SYSACT and change the account numbers on another computer! If you try to use the following programs:

DSKANA
DSKPAK
FORCE
MOUNT
SUBMIT
SYSACT

or any of the certification, video backup, or formatting programs (such as CRT420, VCRSAV, or FMT210) over the network, you will receive an error message. Some other commands also may not work over the network because they would have no logical application over a network.

FILE LOCKING

If you use your own applications programs over the network, be aware that AMOS releases earlier than 2.1 do not provide file or record locking over the network. See Chapter 4 for more information.

GRAPHICS CONVENTIONS

This manual conforms to the other Alpha Micro publications in its use of a standard set of graphics conventions. We hope these graphics simplify our examples and make them easier for you to use. Unless stated otherwise, all examples of commands are assumed to be entered at AMOS command level.

SYMBOL	MEANING
devn:	Device-Name. ***dev*** is the three letter physical device code, and the ***n*** is the logical unit number. Examples of device names are DSK0:, DSK5:, WIN1:, and MTU0:. Usually, device names indicate disk drives, but they can also refer to other devices.
Filespec	File Specification. A file specification identifies a specific file within an account. A complete filespec is made up of the devn:, the filename, the file extension, and the project-programmer number. For example: `DSK0:SYSTEM.INI[1,4]`
[p,pn]	Project-Programmer Number. This abbreviation represents an account on a disk you can store files and data in. An actual disk account number looks like this: [100,2] or [1,4]. Disk account specifications are sometimes referred to as Project-programmer numbers.
{}	Braces are used in some examples to indicate optional elements of a command line. In the example: `DIR{/option}` the braces tell you ***/option*** is not a required portion of the DIR command line.
\	The slash symbol precedes a command line option request. For example: **`DIR/WIDE:3`** RETURN. This command requests a directory display of the disk account you are currently logged into. The option /WIDE:3 indicates you want the display to be three columns wide.
TEXT	**Bold** text in an example of user/computer communication represents the characters you type.
TEXT	Text like this in an example of user/computer communication represents the characters the computer displays on your terminal screen.
text	In a discussion of command syntax, text like this indicates an element you replace with your own entry. For example, ***/option*** tells you to type the appropriate option you want to use.
RETURN	In our examples, the keycap symbol appears whenever you need to press a certain key on your terminal keyboard. The name of the key you need to press appears inside the keycap symbol, like this: RETURN.
	This symbol means **halt!** It indicates an important note you should read carefully before going further in the documentation. Usually, text next to this symbol contains instructions for something you *must* or *must not* do.
	This symbol means **hint**. It indicates a helpful bit of information, or a short cut that could save you time or trouble.
	This symbol means **remember**. It indicates something you should keep in mind while you are following a set of instructions.

CHAPTER 2 —GETTING STARTED WITH ALPHANET

In this chapter, we discuss:

- Finding out the status of your networks
- Getting on and off the network

Once AlphaNET has been installed on your computer, you are ready to put it to use. If you have not yet installed AlphaNET, see the *AlphaNET Installation Guide* for installation instructions.

FINDING OUT THE STATUS OF THE NETWORKS

The NTSTAT program provides you with a status report telling you which networks are available and which computers are currently active on the networks. To see the status display, enter:

```
NTSTAT{/option}
```

/option can be:

/C	to see a continuous, dynamic display of network status
/P	to break a large NTSTAT display into more than one screen page
/F	to display complete list of network parameters
/J	to see what computers are connected to your computer via a virtual terminal connection

These options are discussed in more detail later.

A typical status display looks something like this:

```
Systems Available:

Video Network:                    Status: Node is on the network
  (self)   4261609474-  Accounting                    AMOS  2.2C(452)-13
  SHIP:    4261609475-  Shipping Dept.                AMOS  2.2C(452)-13
  RESEAR:  4261675009-  Research and Development      AMOS  1.4C(452)-13

Network  2:                       Status: Node is on the network
  (self)   33619970-    Accounting                    AMOS  2.2C(452)-13
  SALES:   33619969-    Sales Department              AMOS  2.2C(452)-13
  VP:      33619971-    Vice President's System       AMOS  2.2C(452)-13

Network  3:                       Status: Node is on the network
  (self)   50397184-    Accounting                    AMOS  2.2C(452)-13
  CEO:     50397185-    CEO's System                  PC/AT    DOS Ver 6.2
  MARKET:  50397186-    Marketing                     AMOS 1.4(452)-13

Gateways Available:

Distribution Center             Forward to MARKET:
Government Data Center          Forward to RESEAR:
```

In the display above, your computer is listed first (self). The number, 4261609474-, is your cpuID. **Accounting** is the identifying name assigned to your computer in the SYS:SYSNAM.DAT file. The last display on the line tells you what operating system you are running under. (See the *AlphaNET Installation Guide* for instructions on setting up computer node names and cpuIDs.)

Two other computers, SHIP: (Shipping) and RESEAR: (Research and Development) are also listed. There may be other computers on that network not shown here because they are currently not connected. (See “Turning the Network On and Off,” below.) As you can see, there are other networks, and even gateway connections, also listed in the display.

Each computer shown on the NTSTAT display has a physical connection to the network, but is not necessarily currently active on the network.

If you have a network with a large number of users or if your computer is part of several networks, the network status information displayed by NTSTAT will not fit on a single display screen. To display the network status one page at a time, you can follow the NTSTAT command with the /P option.

If you want to see a continuous, dynamic display of information about the network, you can specify the /C switch with the NTSTAT program. The display will look something like this:

```
                            AlphaNET Status
                              02:15:53 PM
Accounting Network                                  Node count:  6

Status: Node is on the network                 Uptime: 04:00:43
+-----------------------------------------------------------------+
| Block Writes:              22      Block Reads:              34 |
| Network Adds:               4      Network Deletes            5 |
| Transfers of Control:   40668      Transmit Errors:           0 |
| CRC Errors:                 0      Protocol Errors:           0 |
| Lost Packets:               0      Framing Errors:            0 |
| Last Node Added:        HIRES:     Last Node Deleted:     TECH: |
+-----------------------------------------------------------------+
Group  Node      CPU      Ersatz    Group  Node    CPU       Ersatz
-------------------------------------------------------------------
  2     1   1694629889-   (self)      2     9   1694629897-  RODDEV:
  1     2   1694564354-   COMLB2:     2    10   1694629898-  ENGHBF:
  2     2   1694629890-   FS2000:     2    11   1694629899-  DUNC:
```

When using /C, you can use ^T and ^R to page forward and backward in the list of available computers, and [ESC] to switch between different networks.

Network Specific Information

If you follow NTSTAT with a number (e.g., NTSTAT 101), NTSTAT will only display information relating to that particular network. If you specify a network that doesn't exist you get the message:

```
%Network or gateway not found
```

You can also use the /F option to display a complete listing of all network parameters. This is primarily of interest when setting up or debugging a network.

If you use NTSTAT/F on an Ethernet network, it displays the Ethernet addresses associated with each node. Example:

```
Networks:
1694629897-   Network 101  Group   2  Node     9   (AM362 @ 26338)
                                    Ethernet address: 0000B1000001
                           Group   2  Node     4, 1694629892- 02CF1FA00079
                           Group   2  Node     2, 1694629890- 02CF1FA00085
                           Group   2  Node     3, 1694629891- 02CF1FA00081
                           Group   3  Node     5, 1694695429- 02CF1F605520
                           Group   2  Node     8, 1694629896- 0000B100000B
                           Group   3  Node     1, 1694695425- 0000C0081E13
                           Group   3  Node     4, 1694695428- 02CF1F605533
                           Group   2  Node     7, 1694629895- 02CF1FA00084
                           Group   2  Node     1, 1694629889- 02CF1FA00038
                           Group   1  Node     2, 1694564354- 02CF1FA00034

Statistics:

Block Writes:                   2919    Block Reads:          3622
Network Adds:                     18    Network Deletes:         8
Transfers of Control:              0    Protocol Errors:         0
CRC Errors:                       20    Framing Errors:          2
Lost Packets:                      0    Transmit Errors:         18

Local Sockets:

            10 NETSER     No messages pending
            11 VTSER      No messages pending
            14 ANSER      No messages pending
             7 TASK       No messages pending
            16 TSKABB     No messages pending
```

Next you see the same display shown in the discussion of the NTSTAT /J option below.

The address following the text `Ethernet address:` is the Ethernet address for the computer NTSTAT is being run on.

You can use the NTSTAT /J option to see what computers are connected to your computer via a virtual terminal connection. The screen generated by the NTSTAT/J command displays the following information:

- The spawned JOB name.
- The name of the computer for which the spawned job was created.
- The network, group, node, and socket number associated with the computer which made the connection.

The following is an example of the terminal display screen generated by NTSTAT/J:

```
        VIRTUAL TERMINAL SERVICE: CONNECTED JOB PAIR DISPLAY
       ====================================================
Spawned Job    Type ---------- Connected Remote Job Defined By ---------
  job name        System Name   Network #    Group #   Node #   Socket #
 TSKAAA         2   TECH:          5           1        2         4
 TSKAAB         2   SQA:         101           2        5        12
 TSKAAC         1   DEMO2:         1           1        8         9
```

Note that the virtual terminal connection type is shown for Ethernet connections. See Chapter 5 for more information on Type 1 and Type 2 connections.

TURNING THE NETWORK ON AND OFF

The SET program can only be executed from the OPR: account (DSK0:[1,2]). There are two options allowing you to connect and disconnect your computer from a network. The formats are:

`SET LINK` *`cpuID-`*

and:

`SET NOLINK` *`cpuID-`*

cpuID is the identifying number of your computer on the network you wish to turn on or off. For example:

`SET LINK 16842757-` RETURN

You must specify the cpuID, or an ersatz name that includes the cpuID. For Ethernet and SerialNET networks, using SET LINK causes the node to send an “I’m up” message to all other nodes and to complete a list of nodes present on the network.

For Ethernet networks, using MONTST or SET NOLINK causes the node to send an “I’m going off the network” message to other nodes. This reduces lengthy time outs waiting for a `remote not responding` message. The same action occurs for SerialNET networks when SET NOLINK is used (but not when MONTST is used).

In AMOS 2.1 and later software releases, the SET LINK and SET NOLINK commands can only be executed from DSK0:[1,2]. If you are running an AM-340 Ethernet controller on an AM-2000 with an AM-182 cache board, or an AM-3000 computer, you must use SET LINK before turning on the cache via CBEN. Also, once the cache is enabled, you may not turn off the Ethernet network with the SET NOLINK command.

Keep in mind turning off the network disables it for all users on your computer. You will normally want to keep your computer connected to the network at all times; the main use for SET NOLINK is for VideoNET installations which must disconnect from the network before using the video circuitry for backup or restore purposes.

To avoid causing problems for other users on your computer, this operation is best left to a single person, normally that person designated as the System Operator.

VIDEONET AND VCR BACKUP

One of the limitations of the video cable network is your computer *cannot* be logically connected to the network if you are going to do a VCR backup or use the CRT610 program to certify a video tape. This is because the network is using the video circuitry, which cannot run both the network and the VCR at the same time.

Therefore, if you wish to use the VCR software, you must disengage the VideoNET network using SET NOLINK.

You may want to create a command file to perform the steps of your video tape backups or certifications. Make sure the command file contains a SET NOLINK command. You probably want to modify any existing command files also.

Using networks other than VideoNET does not affect your ability to perform video backup functions; only VideoNET must be disconnected.

CHAPTER 3 —USING ALPHANET

In this chapter, we will discuss:

- What can you do on AlphaNET?
- Using ersatz names on AlphaNET
- Protecting devices from access

WHAT CAN YOU DO ON ALPHANET?

The next few chapters will discuss the tasks you can do using AlphaNET. These include:

- Using ersatz names
- Using gateways
- Turning on and off network device access
- Sending messages to users on other computers
- Getting lists of files on other computers
- Copying files to and from other computers
- Printing files across a network
- Connecting your terminal to another computer

USING ERSATZ DEVICE NAMES

Before we get into the uses of the network, we'll explain what ersatz names are, and how you can use them to make AlphaNET easier to use. An ersatz name is a substitute name. Ersatz names allow you to replace a long string of characters with a shorter, easier to remember name. They can save you time and trouble when typing network commands. For example, instead of entering:

```
COPY ACCT.DAT=16842749-DSK12:ACCT.DAT[100,1] RETURN
```

with the properly defined ersatz name, you could enter:

```
COPY ACCT.DAT=PURCH:DSK12:ACCT.DAT[100,1] RETURN
```

See your *AlphaNET Installation Guide* for information on defining ersatz names.

USING GATEWAYS

Network gateways allow multiple AlphaNET networks to be interconnected so all nodes on all networks are available to each other. It does so in a way that makes all nodes appear in the same way. That is, no matter how many different networks are involved, all you need to do is refer to the desired computer by its cpuID or ersatz device name.

PROTECTING DEVICES FROM NETWORK ACCESS

This process should be handled by the System Operator.

If you wish to use the network, but you have files or data you do not wish to allow anybody on another computer access to, you can protect them by closing off access to the device they reside on. Say you have a computer set up with five logical devices. If you don't want people from other computers to access DSK2:, enter:

```
SET NOACCESS DSK2: RETURN
```

People on your own computer will still be able to copy from DSK2:, but nobody can do so across AlphaNET. If you wish to permit access to that device again, enter:

```
SET ACCESS DSK2: RETURN
```

You may also specify an asterisk:

```
SET NOACCESS * RETURN
```

Which means ALL the devices on your computer cannot be accessed across AlphaNET. In this case, your computer can use AlphaNET to access other computers, but your files cannot be accessed by other computers on the network. The DEVTBL command can be used to display the current protection status of your devices.

CHAPTER 4 —ACCESSING FILES OVER ALPHANET

In this chapter, we discuss:

- How files can be accessed across AlphaNET connections
- How to get a directory of another computer's files
- Copying files between computers
- Printing over the network

ACCESSING FILES OVER ALPHANET

AlphaNET has been designed to make network access to files as much like local access as possible. The only exceptions come when a certain operation either requires direct hardware access (such as formatting disks) or is not allowed for security reasons.

In general, any file specification can include a network cpuID, allowing you to specify a file on a remote computer with any command. This means you can copy, edit, update, or otherwise use a remote file just as if it was present on your own computer.

For security reasons, several conditions must be met, however, before access to remote files is allowed. First, SET ACCESS must have been used on the device on which the remote file resides. If a access is allowed for a particular device, and the remote computer is using AMOS 2.0 or later, the file in question must also allow remote access.

Note the default file protection allows read and execute access by remote users, but does not allow writing by remote users. When using the AMOS 2.0 extended directory structure, you can select the protection level you wish to control the level of remote access.

The other important thing to keep in mind is to exercise care when when accessing remote files that may be in use by other users. This is because computers using AMOS releases prior to AMOS 2.1 do not have multi-user file update protection for remote files. See "Remote File/Record Locking Warning for AMOS Releases Prior to 2.1" below for details.

AlphaNET 2.1 *does* provide remote multi-user file protection if the remote users are also using AMOS 2.1 or later operating system releases.

Some programs, such as SYSACT and DSKANA, do not allow themselves to be used across the network. This is to prevent remote users from being able to corrupt a computer or otherwise bypass computer security.

When possible, *pull* files from other computers rather than trying to *push* them. To pull a file, copy a file from a remote computer to your local computer. You push a file when you copy it from your local computer to a remote one. (The best way to accomplish this is to connect to the remote computer via a virtual terminal connection, and pull the file from your local computer.)

Attempts to push files will often result in protection violation errors when write access is denied by the remote computer. This is particularly true when trying to push a file in such a way that you would need to create a new account. This operation is not allowed and, depending on the revision levels of AMOS involved, can cause file system damage. (It is not possible to prevent all occurrences of this without requiring all networked computers to use AMOS 2.0; a restriction we have chosen not to make.)

Here are some things to keep in mind when accessing files across the network:

1. If you are using an operating system earlier than AMOS 2.0, you cannot access files on remote computers running with AMOS 2.0 (or later) operating systems that are configured with extended disk format.

2. With AMOS 2.0 and later software releases, the accounts on the disk drive are not always in sequence; account [1,4] may appear after account [44,44]. This condition may cause problems when accessing files across the network.

   For example: If your local computer is running AMOSL 1.3D software, you will not be able to access files over the network on computers running AMOS 2.0 (or later) with a traditional file structure, if the accounts on the disk you are accessing are not in sequence. The MFDSEQ program can be used to sequence the account numbers on computers running AMOS 2.0 (or later) software with traditional (non-extended) file structures.

3. You will not be able to access files on a remote computer that must be accessed through a gateway, if the files on the remote computer are located on a device that is not defined on your local computer.

   For example: If the files you wish to access on the remote computer are located on ESD0: and you do not have an ESD0: device defined on your local device, you will not be able to access the files.

Remote File/Record Locking Warning for AMOS Releases Prior to 2.1

When using versions of AMOS earlier than 2.1, there are some restrictions that must be observed when using either LOKSER or AMOS 2.0's integrated locking facility. While these restrictions

have been removed in AMOS 2.1 (and later) releases, those computers using AMOS releases prior to 2.1 must be aware of the restrictions outlined in the following paragraph.

FILE LOCKING WARNING

Files being accessed by users on the same computer as the file (local access) are protected from improper access by remote users. However, two remote users, from one or more remote nodes, are not protected from each other. Furthermore, if a remote user opens a file, but does not close it, either because of program error or computer crash, the file will remain locked indefinitely. It is then necessary to use LOKUTL to remove the file lock so other users may access the file. For more information about file and record locking see the *AMOS File Locking User's Guide*.

As of AMOS 2.1, file locking for multiple remote users is handled automatically.

GETTING A LIST OF FILES ON ANOTHER COMPUTER

To find out what files are available on a remote computer, use the DIR command across the network just as you would on a local computer. Just include the cpuID (or a valid ersatz name) in the normal file specification:

```
DIR 12930293-DSK3:[123,6]*.TXT RETURN
```

or:

```
DIR LEDGER:DSK0:[1,4] RETURN
```

The wildcard specification **ALL:** is not supported over the network.

COPYING FILES BETWEEN COMPUTERS

You can use the standard COPY program to copy files across the network. The only difference is you need to specify the cpuID. The format is:

```
COPY {full-filespec}={cpuID-}source-full-filespec
```

For example:

```
COPY ACCT1.TXT[50,2]=16844425-DSK0:ACTTMP.TXT[100,5] RETURN
COPY WRHREC.DAT=16842756-DSK2:WRHREC.DAT[10,5] RETURN
```

If you use ersatz device names, you must not follow the colon with a space. The standard AMOS conventions of wildcarding and file specifications apply.

Copying files between computers connected by gateways works exactly the same way.

PRINTING OVER THE NETWORK

AlphaNET allows you to send a file to another computer for printing.

You may do so using either the Task Manager print spooler, or the memory based print spooler (LPTSPL).

Printing Using the Task Manager Spooler

The format for the task manager printer spooler is:

```
PRNT{/option} {cpuID-}{printerspec}={filespec{,... filespec(s)}}{/option}
```

Some examples:

```
PRNT 16842758-LPT1=PAYROL.LST RETURN
PRNT GNLDGR:DIABLO=RESULTS RETURN
PRNT 14932034-LASWRT=MASTER.PAY/C:4 RETURN
```

If you use a printer on another computer regularly, you will probably want to set up a command file to make it easier to send files to the printer.

Because of the procedure PRNT uses to select defaults, the specification can be a little tricky when the computer you want to print to has different device names than your own computer. If the computer you are sending to does not have a device name defined on it with the same device name as you are logged into, you will have to specify a device name that does exist on that remote computer (the safest bet is DSK0:, which all computers must have) on the left side of the equal sign.

For example, say you are logged into DSK5:[100,0] and you wanted to send a file to a printer on another computer that does not have a DSK5:. If you specify:

```
PRNT 16842758-LASWRT=REPORT.LST RETURN
```

PRNT assumes the output device is the same as the input device (DSK5:, where you are logged into). When it looks for a DSK5: on the other computer, it doesn't find it, and stops, giving you an error message:

```
?Cannot INIT DSK5: - device does not exist
```

If you see such a message, try again, specifying a valid device name on the printer's computer on the left side of the equal sign. Or you may try logging into DSK0: on your computer, and making the print request from there.

As with any AMOS command specification, the PRNT command must be able to access the printer, and the place where the file to be printed resides. Therefore, if the disk you want to print from is set to NO ACCESS, or the output disk specified on the printer computer is set to NO ACCESS, you will also see the above error message.

You may optionally place a file called PRINTR.DFT in account [1,4] of your system disk. This file contains a cpuID which the network will use as a default on all PRNT requests. If this file does not exist, or contains a 0, then PRNT will default to your own computer. This can be useful if the printer you use most often is on another computer.

In order to use AlphaWRITE and AlphaCALC to print to a remote printer, you must have version 1.2 or later of these products.

Printing Using the Memory Based Spooler (LPTSPL)

You can set up the memory based print spooler to allow remote printing, even if your local computer is using the task manager spooler.

On each computer on which you wish to have access to a remote printer, create an .INI file (of whatever name you like) in DSK0:[1,4] which contains the following:

```
REMOTE = cpuID
NAME = name
FORMS = NORMAL
```

cpuID is the full cpuID number (ersatz names are not allowed here) of the computer the printer is attached to, and ***name*** is the name of the printer as defined in the printer initialization file on that remote computer. You may also include the DEFAULT command if you wish. All other commands will be ignored.

Be careful when choosing a name for the printer in the printer initialization file. Make sure it is a *unique* name compared to all the other network-accessible printers or any printers defined on your local computer.

Once you have created the initialization file, you will need to add an LPTINI statement to your system initialization command file to define the printer each time you boot the computer. Unlike when setting up a local printer, you do not need a separate job to execute the LPTINI command. Instead, simply execute the LPTINI command directly in the system initialization command file. The LPTINI command will define the printer and exit, continuing with the remainder of the initialization file.

The LPTSPL spooler has two restrictions when used with remote printers:

1. You cannot display a list of a remote printer's queue.

2. You can not use PRINT across a network using an ersatz name in place of the cpuID number.

Make sure the remote printers you define using the LPTINI command are really printers using the memory based spooler.

CHAPTER 5 —CONNECTING TO A REMOTE COMPUTER

In this chapter, we discuss:

- How to connect your terminal to a remote computer
- Virtual terminals
- Virtual terminal drivers
- Finding remote computer connections

AlphaNET supports a virtual terminal capability which allows your terminal to appear to be directly connected to a remote computer. This lets you do just about any operation you could perform locally.

CONNECTING TO A REMOTE COMPUTER—CONECT COMMAND

Use CONECT to connect your terminal to a remote computer. The command syntax is:

```
CONECT{/S} cpuID- {memory{K}{M}}
```

cpuID is the cpuID or ersatz name of the computer you want to connect to. ***memory*** is an optional amount of memory to allocate on the remote computer. You can enter the memory amount in number of bytes (i.e., 65535), or with a K (kilobytes) or an M (megabytes). If no memory size is specified, a default of 64K will be used. Typing CONECT with no arguments will display help information.

The optional /S tells the host computer to send the characters to the terminal more slowly than usual. One example of when you might wish to choose slower terminal output speed is when a local printer is connected to a terminal, and you have the terminal set to transparent print mode to cause all output to go directly to the printer. The printer cannot process the incoming characters as quickly as a terminal would.

If the remote computer you are attempting to connect to has a system password, you will be asked for that password. If your computer makes use of user names, CONECT will attempt to log you in on the remote computer under your current user name. If no such user name exists on that remote computer, you will need to log yourself in.

You can only connect to a computer running VTSER. To find out what computers are available for virtual terminal connection, use the FNDVT program described below. For information on VTSER setup, see the *AlphaNET Installation Guide*.

Once a connection has been established, you can use it as you would any other terminal connection.

User License Restriction for Ethernet Type 2 Connections

If your Ethernet Type 2 connection to a remote computer would exceed the number of Type 2 virtual terminal connections allowed for in your AlphaNET user license, the connection will be converted to a Type 1 connection. For information on determining how many connections your host computer is licensed for, see the information on NETINI/P in the *AlphaNET Installation Guide.*

Disconnecting from the Remote Computer

When you are done using the remote connection, you can disconnect by entering this key sequence: Press | (broken vertical line character). Release, then press D.

If for some reason there is a crash on the remote computer you are connected to, there will be a delay of approximately 60 seconds, after which you will be disconnected and returned to your local computer.

If you should need to enter a “|”character while using your terminal connected to a remote computer, enter “||” (this way, it is not confused as a CONECT command key sequence).

Displaying the Connection Path

Because you can use the CONECT program multiple times to chain through a series of computers, you can press | then release and press P to display your current connection path.

NOTES ON REMOTE CONNECTIONS

The following sections contain hints on using the virtual terminal connection on a remote computer.

Using MULTI on a Remote Computer

After you use CONECT to make a virtual terminal connection to a remote computer, you can execute a large number of AMOS programs, including MULTI, the AMOS environment manager. However, the virtual terminal program will not operate properly if you attempt to use CONECT to connect to a third computer while running MULTI. CONECT will allow you to make the third connection, but when you disconnect, you will end up at your local computer instead of the remote computer where you were running MULTI. This leaves the spawned job on the remote computer open; it can only be terminated by rebooting the remote computer.

Performance Enhancements for Ethernet Networks

The performance of the virtual terminal connection is directly related to the performance of the underlying network connection. The faster the network, the better the performance. You will also see an effect based on the number of simultaneous virtual terminal connections.

The usual effect of a slow network connection will be a delay in echoing individual characters as they are entered from the keyboard. Terminal output performance is usually quite good even over a very slow network connection.

SerialNET at 19200 baud or faster gives good virtual terminal performance. VideoNET is slow, but usable.

The best possible virtual terminal performance can be achieved by users who meet the following criteria:

- Running AMOS 2.2 or later operating systems.
- Running AlphaNET 2.2 or later.
- Using Ethernet as your data transport mechanism

Users who fall into the above category will enjoy a data transfer rate that is significantly faster than the performance found in earlier AlphaNET releases because they will be able to use Type 2 connections (see the section below). Your computer will be able to support a larger number of virtual terminal connections with much greater efficiency. You will also be able to use the command line editor for remote connections.

To use these new features under CONECT, you will need to add a special TRMDEF statement. You must also create a special communication zone, which requires one additional statement in your system initialization command file. See the *AlphaNET Installation Guide* for information on adding the special TRMDEF statement and communication zone.

Benefit of Type 2 Connections

When you make a virtual terminal connection using Ethernet to a host computer running AlphaNET 2.2 or later and AMOS 2.2 or later, you see the message `Virtual Terminal Connection Type 2`. Virtual terminal connections made with earlier versions of AlphaNET or AMOS, or using network media other than Ethernet, will display `Virtual Terminal Connection Type 1`.

Type 2 connections provide a significant performance enhancement to your network, and also allow the use of the AMOS line editor and 8-bit character support. Why are Type 2 connections faster? With Type 1 connections, all communication between the connecting terminal and the host computer go through a single job, VTSER (virtual terminal service). This makes VTSER a bottleneck when multiple connections are in effect, slowing down communication. With Type 2 connections, most of the communication takes place directly between the connecting terminal and its spawned job on the host, freeing up VTSER for improved performance, and decreasing overhead.

Terminal Drivers for Virtual Terminals

The virtual terminal facility does not do terminal emulation or translation, but instead establishes a virtual circuit between the remote computer and your terminal. This eliminates a source of problems with terminal and function key compatibility.

The AMOS line editor is not supported when you are attached as a virtual terminal on a remote computer unless you are using Ethernet, AlphaNET 2.2 (or later), and AMOS 2.2 (or later).

In order to establish a virtual terminal connection, the terminal driver in use by your physical terminal must be available and installed on the remote computer. If it isn't, an error message is displayed and the connection refused.

You can ensure a terminal driver is available by defining a dummy terminal using it (for pre-2.0A AMOS computers) or by using the TDVDEF program (under AMOS 2.0A and later). Using the TDVDEF program is described in the *AMOS System Commands Reference Manual.*

Virtual Terminals and 8-Bit Characters

If you want to execute an application that uses 8-bit characters over a virtual connection and you are not using an Alpha Micro terminal, it may be necessary to create a special terminal driver for use while connected to a remote computer.

The following information is provided to help you determine whether or not you will need to create this special terminal driver for the remote computer. This information is directed at software developers and assumes a familiarity with assembly language programming and the AMOS Terminal Service System (Reference *AMOS Monitor Calls Manual*, Chapter 7).

When in 8-bit mode (T$EXT is set) a function key is expected to consist of a two-character sequence, which is a CSI character (233 octal) followed by the function key character. Alpha Micro's 8-bit terminals have been programmed to generate this two character sequence.

While connected to a remote computer, each key pressed on the physical terminal passes through the terminal driver and the AMOS terminal service input routines on the local computer. Then it is passed on to the remote computer where it passes through the comparable routines on the remote computer.

The CONECT program puts the local terminal in the following modes in an attempt to preserve the input stream as it comes from the terminal:

T$DAT - data mode
T$XLT - let the application translate function keys
T$EXT - 8 bit character support

With these modes set, if you are using an Alpha Micro terminal, the data received from the terminal will be passed on to the remote computer exactly as it is received. Consequently, Alpha Micro terminal drivers will function properly on either the local computer or the remote computer.

On the other hand, if your terminal does not generate the proper two-character sequence and consequently your terminal driver performs the necessary conversion, the input stream that is passed to the remote computer will not be identical to that received from the physical terminal. In this case the terminal driver on the remote computer will need to be modified to look for this altered character sequence.

Restrictions for Virtual Terminals

Certain programs may not be used when connected by virtual terminal. These are programs which use the AMOS JLOCK monitor call, locking out all other jobs. Because the VTSER job must be able to run in order to service the virtual terminal connection, executing a program which performs a JLOCK will result in the virtual terminal connection being terminated.

The following programs are known to perform JLOCKs and cannot be used over a virtual terminal connection:

- VCRSAV, without the /T option.
- STRSAV.
- Any program during its PIC installation procedure.

The AMOS 2.0 BACKUP program *will* work correctly over a virtual terminal connection (except, of course, for video backup without /T).

Some programs may not work because of the difference in operating environments between a normal direct connection and a remote connection. For example, if you were to execute the BAUD command to change the baud rate of your virtual terminal, the baud rate of your direct connection to your terminal would not change.

COMMAND FILE NETWORK DISCONNECT—DISC COMMAND

For those cases when you wish to automatically disconnect from a remote computer when a task is finished, include the DISC command at the end of a command file to be processed on the remote computer. When DISC appears as the last line in a command file, when AMOS finishes processing the command file on the remote computer, it will automatically disconnect your virtual terminal connection.

LOCATING COMPUTERS AVAILABLE FOR CONNECTION

The FNDVT program is used to locate remote computers which can be connected to by a virtual terminal. Enter the FNDVT command at AMOS command level:

FNDVT RETURN

After a brief pause, you will see a display of the available computers, including their ersatz device name or cpuID, their symbolic name, the number of active jobs currently on the computer, and the uptime of the computer. For example:

```
Systems available for remote connection:

ACCNT:  Accounting         126 active jobs, uptime 342:25:57
SALES:  Sales Computer      15 active jobs, uptime 24:46:21
INVEN:  Inventory System     7 active jobs, uptime 97:12:56
```

FNDVT will not find and display computers that must be accessed through gateways.

CHAPTER 6 —MISCELLANEOUS APHANET COMMANDS

In this chapter, we discuss:

- How to send messages to remote computers
- How to get status information about a remote computer
- Support for non-AlphaNET protocols
- How to measure the speed of a network connection

SENDING MESSAGES TO OTHER COMPUTERS

To send a message to someone on another computer, use the normal SEND command, but add a cpuID (or ersatz device name). The format is:

```
SEND cpuID-jobname message
```

For example:

```
SEND 16842758-JESSICA WHAT DID YOU WANT? RETURN
SEND SALES:ROGER NEED TO TALK TO YOU ABOUT EDDIE RETURN
```

If you use ersatz device names in SEND, do not put a space after the colon.

What do you do when you want to send a message to someone over the network, but you're not sure what his or her job name is? Use the RJOB command, which will display the job names for each user on the remote computer. RJOB is described later in this chapter.

GETTING STATUS INFORMATION FOR REMOTE COMPUTERS

When using a network of several computers, you will often want to determine the status of a remote computer, perhaps before attempting some special operation. Two programs, RSYS and RJOB, are provided to display such status information.

RSYS—Remote Computer Status

The RSYS program displays a brief status message on a specified computer. It shows the number of active jobs on the computer, and the total system uptime. This command is a quick way of determining if a specific computer is up and available. To use RSYS, follow it with a cpuID or valid ersatz device name:

```
RSYS 16842759- [RETURN]
RSYS ARCHIV: [RETURN]
```

RSYS will respond with a message such as: `126 active jobs, uptime 628:37:41`. If the specified remote computer is not available (it is not up or is not on the network), you will see a message to that effect.

RJOB—Remote Job Status

The RJOB program displays a brief list of active jobs on a remote computer. It displays the job and terminal name, the user name, and the program being used. To use RJOB, follow it with a cpuID or valid ersatz device name:

```
RJOB 436842359- [RETURN]
RJOB ACCNT: [RETURN]
```

RJOB will respond with a display such as:

```
ACCT    JOB1    Accounting          DIR
TASK    MANAGR  System Service      TSKIDL
NETSER  NETSER  Network Service     NETSER
VTSER   VTSER   Network Service     VTSER
```

If the specified remote computer is not available (it is not up or is not on the network), you will see a message to that effect.

SUPPORT FOR NODES USING NON-ALPHANET PROTOCOLS

The following commands improve performance for computers not running the AlphaNET protocol:

```
SET NDV cpuID
SET NONDV cpuID
```

These commands are used to suppress network “keep alive” messages that are not used by non-AlphaNET protocols, thus reducing network overhead.

SUPPRESSING ETHERNET IDENTIFICATION BROADCAST

Your local computer periodically broadcasts identification packets to let other nodes on the Ethernet network know you are active on the network. Under certain circumstances, the normal broadcast of node IDs that occur on a network can cause problems (for example, when nodes are tied together by modems, and each ID broadcast triggers phone-line charges). To temporarily turn off this broadcast, use the SET NOBROADCAST command; to turn it back on, use SET BROADCAST:

```
SET BROADCAST cpuID
SET NOBROADCAST cpuID
```

where ***cpuID*** specifies your node.

DELETING A SPAWNED JOB

When a virtual terminal connection is abruptly discontinued without having a chance to clean up after itself (for example, the local computer "crashes"), a spawned job is left open on the remote computer. The KTASK command allows you to clear up spawned jobs left on your local computer. To use KTASK, you must log into OPR:. Its format is:

```
KTASK jobname
```

where ***jobname*** specifies the spawned job. Spawned job names take the form of TSK??? (for example, TSKAAB). KTASK may also be used to clean up spawned jobs left by AlphaTCP. Be careful not to delete a spawned job that is still in use.

MEASURING NETWORK CONNECTION SPEED

When using a collection of networks with gateways interconnecting them, it is sometimes useful to find out the relative speed of different connections. The NETSPD program measures and reports on the actual round-trip time needed to send a packet of information to a specified computer. NETSPD has one option, NETSPD/L.

NETSPD displays the minimum, maximum, and average round-trip times, based on a sample of ten separate packets. The time displayed is expressed in milliseconds.

NETSPD will display a message such as:

```
Measured round trip packet transit time:
Min = 43 ms, Max = 45 ms, Avg = 43.8 ms.
```

If the specified remote computer is not available (not up or not on the network), it will tell you so.

To use NETSPD, follow it with a cpuID or valid ersatz device name:

```
NETSPD 16842759- RETURN
NETSPD ARCHIV: RETURN
```

NETSPD/L is a more accurate and comprehensive way of measuring your network performance. The /L option allows you to send a 900 byte data packet of uncompressable data. You can also specify the number of packets you wish to send. If you do not specify the number of packets to be sent, the program will send the default quantity, 10 packets.

To use NETSPD/L, follow it with a cpuID (or valid ersatz device name) and the number of packets you wish to send:

```
NETSPD/L 16842759- 100 RETURN
NETSPD/L ARCHIV: 100 RETURN
```

NETSPD has an error message that will warn the user when a packet is sent to a remote computer and the packet echoed back is not the same packet. The error message reads:

```
?Sequence error. Expected n received x
```

n is the sequence number of the packet sent out and ***x*** is the sequence number of the packet received. This type of error gets generated when the network effectively gets out of sync and gets the packet order incorrect.

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